



“Explain computer vision applications in agriculture, identifying practical problems and presenting solutions using structured tools and visuals”

COURSE NAME AND CODE: ITA0518-COMPUTER VISION

FACULTY: Dr. T. M. USHA

NAME: HARITHI.S(192421151)

Introduction & Need

What is Computer Vision?

- AI technology that analyses images and videos.
- Helps computers identify crops, diseases, and objects.

Why Agriculture Needs It

- Detects crop diseases early.
- Reduces manual labour.
- Improves crop yield.
- Saves water and farming costs.

Applications

Major Applications

- 🌱 **Crop Disease Detection** – Identifies leaf diseases using AI.
- 🌿 **Weed Detection** – Detects weeds for targeted herbicide spraying.
- 🍏 **Fruit Detection** – Identifies fruit ripeness for harvesting.
- 💧 **Smart Irrigation** – Monitors crop health to optimise water usage.



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Problems & Solutions

Problem

- Crop diseases
- Weed growth
- Water wastage
- Labor shortage

Computer Vision Solution

- AI-based leaf image detection
- Automatic weed identification
- Smart irrigation monitoring
- Automated harvesting systems

Tools Used:

- OpenCV
- TensorFlow
- Python
- YOLO (Object Detection)
- Drones & Smart Cameras

Benefits & Challenges

Benefits

- ✓ Early disease detection
- ✓ Higher crop productivity
- ✓ Reduced labour costs
- ✓ Water conservation
- ✓ Precision farming

Challenges

- High initial cost
- Requires quality image data
- Internet and technical support needed

Conclusion

Computer Vision is transforming agriculture by improving crop monitoring, disease detection, irrigation, and harvesting. It helps farmers make better decisions, increase productivity, reduce costs, and promote sustainable farming.

Future Scope

- AI-powered farming robots
- Drone-based crop monitoring
- IoT-enabled smart agriculture
- Real-time pest and disease prediction



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Thank
You